

Vinay Pokharkar

AI Full-Stack Developer

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Summary

- AI Full-Stack Developer building production LLM applications, RAG pipelines, and agentic workflows end to end on FastAPI, Next.js, and React Native. Shipped for a paying freelance client and an early-stage startup, placed Top 12 of ~6,000 at the Sarvam AI Builders Hackathon, and authored merged PRs into open-source projects with 52k+ and 16k+ GitHub stars.

Education

SIES Graduate School of Technology, Nerul, BE in Artificial Intelligence and Machine Learning 2022 - 2026

- GPA: 7.74/10.00
- **Coursework:** Data Structures & Algorithms, Database Management Systems, Computer Architecture, Machine Learning, Deep Learning, Image and Video Processing, Cloud Computing

Technical Skills

AI/ML: LLMs (Gemini, Ollama), RAG, agentic workflows, prompt engineering, vector databases (Pinecone), TensorFlow, Hugging Face, Sarvam.ai, n8n

Languages: Python, TypeScript, JavaScript, SQL

Backend: FastAPI, Node.js, Express, Flask, REST APIs, JWT authentication, Docker

Frontend & Mobile: Next.js, React, React Native, Tailwind CSS, Streamlit

Databases: PostgreSQL, MongoDB, MySQL, Supabase

Cloud & DevOps: AWS, Vercel, Git, GitHub, GitLab, CI/CD

Experience

Software Engineering Intern, HeadRest - Remote April 2026 - July 2026

- Build cross-platform mobile features for an early-stage consumer app from a shared Flutter codebase, shipping Android builds and native iOS home-screen widgets.
- Profiled and reworked the pipeline's slowest backend path, story search and scoring, cutting its p50 latency ~46% (45s to 24s) by migrating to an Exa + compact-scoring backend.
- Design AI-driven automation workflows for content and short-form video generation, owning engineering and product tasks end to end on a fast-moving startup team.

Web Developer Intern, Pioneer Machines & Automation Pvt. Ltd. - Remote June 2025 - September 2025

- Owned frontend and backend API development for an internal web platform, building React UI components and Express REST endpoints backed by PostgreSQL under a feature-based modular architecture.
- Designed JWT-based authentication and role-based access control (RBAC) across admin and standard-user tiers, securing protected API routes.
- Shipped features across schema, API, and UI layers, reviewing and merging PRs via Git throughout the 4-month remote internship.

Certifications

- **Perform Foundational Data, ML, and AI Tasks in Google Cloud** - Google, 2023
- **Supervised Machine Learning: Regression and Classification** - Stanford University (Andrew Ng, DeepLearning.AI), 2024
- **AWS Academy Graduate - AWS Academy Cloud Foundations** - Amazon Web Services, 2025

Projects

Praman - Multilingual WhatsApp banking assistant 2026

- Built a WhatsApp banking and loan assistant for low-literacy rural users, taking voice or text and replying in kind in the user's own language, on Sarvam.ai speech-to-text, translation, and text-to-speech (22 Indian languages), plus a dashboard for tracking conversations and applications.
- Shipped a document explainer: users forward a scanned English bank letter or loan agreement (PDF or image) and get it summarized and read back in their chosen language. Placed **Top 12** at the Sarvam AI Builders Hackathon, Bengaluru (2026), among ~100 on-site finalists picked from ~6,000 registrations.

CodeMotion - Text-to-animation AI platform Github

- Engineered a full-stack text-to-animation LLM application that converts natural-language prompts into Manim-rendered math videos across 3 user tiers (10s, 30s, and 60s outputs), using Gemini for prompt-to-Manim Python script generation.
- Isolated untrusted, model-generated Python inside a Dockerized sandbox to contain execution; integrated a Next.js chat UI, FastAPI orchestration layer, and Supabase for authentication and video storage.

SeaGuard - AI mobile assistant for fishermen (Hackcelestial 2.0 winner) Github

- Co-led a 5-person team to ship a launch-ready React Native mobile assistant targeting India's coastal fishing population, featuring TensorFlow-based Potential Fishing Zone (PFZ) detection, SOS alerts, GPS navigation, catch logging, trip planning, and live weather.
- Built the FastAPI + Supabase backend with Sarvam.ai-powered multilingual voice support and LLM-driven assistance; queued the app for public store submission. Awarded *Most Product Readiness*, Hackcelestial 2.0, PCOE (Sept 2025), from 400+ participants.

AutoPharmX - Cancer drug-efficacy prediction Github

- Built an ML system predicting cancer drug efficacy (IC50 values), spanning data preprocessing, feature engineering, model training, and evaluation in Python.
- Deployed the trained model as a Dockerized FastAPI inference service behind an interactive Streamlit UI, letting researchers query predictions on new compound inputs.

Finsee - Voice-first banking for visually impaired users Github

- Built a fully voice-driven React Native mobile app enabling visually impaired users to perform screen-free digital transactions; owned voice state management, speech recognition integration, and secure API orchestration from UI down to the API layer.

IIMSC - Multilingual recruitment platform (*Freelance*) Live

- Delivered a production multilingual recruitment platform for IIMSC as a paid freelance engagement; built separate employer and candidate portals in Next.js + Tailwind CSS with i18n across 4 languages and industry-specific landing pages, deployed to client-managed cPanel infrastructure.

Open Source

Impeccable (52k+ GitHub stars) - Design skill library for AI coding harnesses Merged PRs

- Authored 4 merged PRs into pbakaus/impeccable (2026): a new platform axis (web / iOS / Android / adaptive) threaded through the skill registry behind a backward-compatible default, Qoder as the 13th harness target, two detector anti-pattern rules, and a Windows-only Chrome GPU crash-loop fix.

Maxun (16k+ GitHub stars) - Open-source no-code web scraping platform Merged PR

- Extended the Website-to-API feature in getmaxun/maxun with REST endpoints for robot run schedules, moving cron-building and timezone validation into a shared helper so API and dashboard paths stay in sync (2026).